Year 13 Further Maths - Circular motion

Similar questions - Vertical circles, rods, hemispheres and cones

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle P of mass 1.5 kg is attached to one end of a light inextensible string of length 0.9 m. The other end is attached to a fixed point O. P is held vertically below O and projected horizontally with speed 7 $\mathrm{m\,s^{-1}}$. When the string makes an angle θ with the downward vertical, the speed of P is v $\mathrm{m\,s^{-1}}$.

(a) Show that $v^2 = 49 - 17.64(1 - \cos\theta)$. **[3]**

(b) Find the tension in the string, and the tangential acceleration of P, when $\theta = 60°$. **[5]**

(c) Find the least speed of projection for which P completes vertical circles. **[3]**

(d) The string breaks when the tension exceeds 60 N. Find the angle θ at which the string breaks for projection speed 7 $\mathrm{m\,s^{-1}}$. **[4]**

2 A particle P of mass 0.8 kg is attached to one end of a light rod of length 1.5 m which is free to rotate about a fixed point O in a vertical plane. P is projected horizontally from the lowest point with speed 9 $\mathrm{m\,s^{-1}}$.

(a) Find the speed of P at the highest point, and the force in the rod there, stating whether it is tension or thrust. **[5]**

(b) Find the least speed of projection for which P completes vertical circles, and explain how this differs for a string. **[3]**

(c) P is now moving in a horizontal circle with the rod making a constant angle of 40° with the downward vertical. Find the speed of P and the force in the rod. **[6]**

3 A smooth solid hemisphere of radius r is fixed with its plane face on horizontal ground and centre O. A particle P is released from rest on the surface where OP makes an angle α with the upward vertical, $\cos\alpha = \frac{3}{5}$.

(a) Show that P leaves the surface when $\cos\theta = \frac{2}{5}$, where θ is the angle between OP and the upward vertical. **[6]**

(b) Find, in terms of r, the horizontal distance from O of the point where P first hits the ground. Give your answer to 3 s.f. **[8]**

4 A particle P of mass 3 kg hangs at rest on the end of a light inextensible string of length 0.6 m, the other end being fixed at O. P is given a horizontal impulse of magnitude 15 N s.

(a) Show that P begins to move with speed 5 $\mathrm{m\,s^{-1}}$. **[1]**

(b) Show that the string becomes slack when it makes an angle θ with the downward vertical where $\cos\theta = -0.75$ (to 2 s.f.) and find the angle the string then makes with the horizontal. **[7]**

(c) Find the speed of P at this instant. **[2]**

5 A smooth hollow cone is fixed with its axis vertical and its vertex downwards. The semi-vertical angle of the cone is 30°. A particle P of mass 0.2 kg moves in a horizontal circle on the inner surface of the cone at height 0.5 m above the vertex.

(a) Show that the speed of P is $\sqrt{gh}$, where h is its height above the vertex, and find the speed. **[6]**

(b) Find the time for one revolution. **[2]**

(c) An identical particle Q rests at the vertex. Taking the vertex as the zero of potential energy, show that the total mechanical energy of P exceeds that of Q by $\frac{3}{2}mgh$. **[3]**

Question 1

(a) Energy: $\frac{1}{2}mv^2 = \frac{1}{2}m(49) - mg(0.9)(1-\cos\theta)$, so $v^2 = 49 - 17.64(1-\cos\theta)$

(b) $v^2 = 49 - 8.82 = 40.18$. Radially: $T - mg\cos 60^\circ = \dfrac{mv^2}{0.9}$, $T = 1.5\,(44.64 + 4.9) = 74.3$ N. Tangential acceleration $= g\sin 60^\circ = 8.49\ \mathrm{m\,s^{-2}}$ (decelerating)

(c) At the top $T \geqslant 0$: $v^2 \geqslant gl$; with $u^2 = v^2 + 4gl$, $u^2 \geqslant 5gl = 44.1$, $u \geqslant 6.64\ \mathrm{m\,s^{-1}}$

(d) $T = m\left[\frac{u^2}{l} - 2g + 3g\cos\theta\right] = 60$: $40 = 54.44 - 19.6 + 29.4\cos\theta$, $\cos\theta = 0.1754$, $\theta = 79.9^\circ$

Question 2

(a) $v^2 = 81 - 4(9.8)(1.5) = 22.2$, $v = 4.71\ \mathrm{m\,s^{-1}}$. At the top: $T + mg = \frac{mv^2}{l}$, $T = 0.8(14.8 - 9.8) = 4.0$ N tension

(b) A rod can provide thrust, so only need $v \geqslant 0$ at the top: $u^2 \geqslant 4gl = 58.8$, $u \geqslant 7.67\ \mathrm{m\,s^{-1}}$ (a string needs $u^2 \geqslant 5gl$)

(c) $T\cos 40^\circ = mg$, $T = \frac{7.84}{\cos 40^\circ} = 10.2$ N tension; $T\sin 40^\circ = \frac{mv^2}{1.5\sin 40^\circ}$, $v^2 = g(1.5)\sin 40^\circ \tan 40^\circ = 7.93$, $v = 2.82\ \mathrm{m\,s^{-1}}$

Question 3

(a) Energy: $v^2 = 2gr(\cos\alpha - \cos\theta)$. Leaves when $R = 0$: $mg\cos\theta = \frac{mv^2}{r}$, so $\cos\theta = 2\left(\frac{3}{5} - \cos\theta\right)$, $3\cos\theta = \frac{6}{5}$, $\cos\theta = \frac{2}{5}$

(b) At leaving: $v^2 = gr\cos\theta = 0.4gr$; horizontal speed $v\cos\theta$ and downward speed $v\sin\theta$ ($\sin\theta = \frac{\sqrt{21}}{5}$), height $0.4r$, horizontal position $r\sin\theta = 0.9165r$. Solving $0.4r = v\sin\theta\, t + \frac{1}{2}gt^2$ for t and adding $v\cos\theta\, t$ gives distance $\approx 1.04r$

Question 4

(a) $I = mv$, $v = \frac{15}{3} = 5$

(b) $T = 0$: $\frac{mv^2}{l}$ equation gives $\cos\theta = \dfrac{2g - u^2/l}{3g} = \dfrac{19.6 - 41.67}{29.4} = -0.75$. $\theta = 138.6^\circ$, so the string makes 48.6° above the horizontal

(c) $v^2 = 25 - 2(9.8)(0.6)(1.75) = 4.41$, $v = 2.1\ \mathrm{m\,s^{-1}}$

Question 5

(a) The normal reaction R is perpendicular to the surface, so makes 30° with the horizontal. Vertically $R\sin 30^\circ = mg$; horizontally $R\cos 30^\circ = \frac{mv^2}{r}$. Dividing: $v^2 = gr\cot 30^\circ$ and $r = h\tan 30^\circ$, so $v^2 = gh$, $v = \sqrt{9.8 \times 0.5} = 2.21\ \mathrm{m\,s^{-1}}$

(b) $r = 0.5\tan 30^\circ = 0.2887$; time $= \frac{2\pi r}{v} = 0.820$ s

(c) $E_P = \frac{1}{2}mv^2 + mgh = \frac{1}{2}mgh + mgh$; $E_Q = 0$ (at rest at the vertex), so the difference is $\frac{3}{2}mgh = 1.47$ J